



github.com/dohagamal693-cyber/Clinic-Management-System

Contact

Clinic Management System

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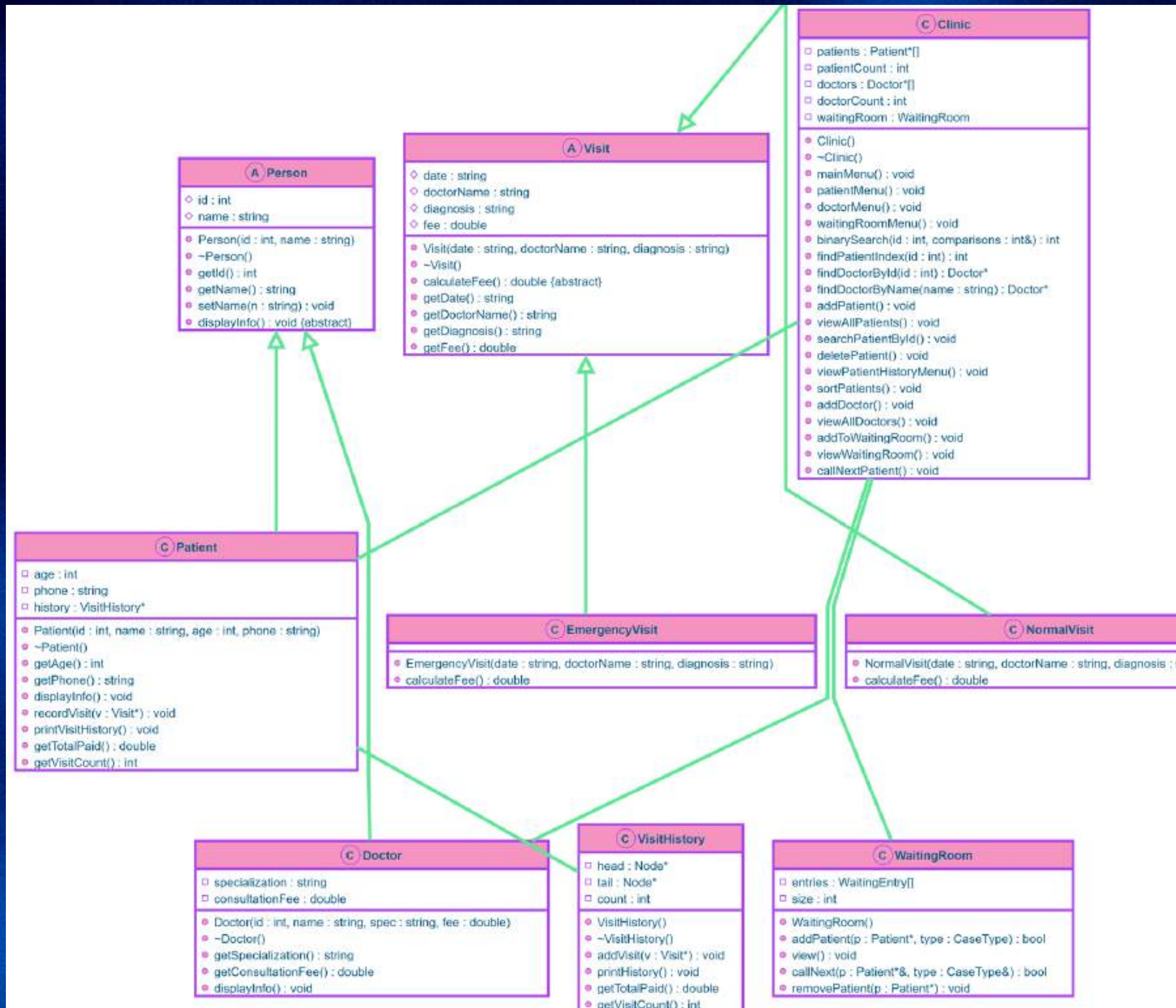
What the Project Does?

- Stores patients and doctors
- Manages today's waiting room using a priority queue
 - Records each visit
- Emergency cases are prioritized over normal cases





UML Class Diagram



OOP in Code

```
class EmergencyVisit : public Visit {  
    public:  
    EmergencyVisit(string date, string  
        doctorName, string diagnosis)  
        : Visit(date, doctorName,  
            diagnosis) {  
        fee = calculateFee();  
    }  
  
    double calculateFee() const  
    override { return 400.0; }  
};
```

- Inheritance: Patient and Doctor inherit from Person

- Polymorphism:
EmergencyVisit and NormalVisit override
calculateFee()

- Emergency visit costs 400 EGP,
Normal visit costs 200 EGP

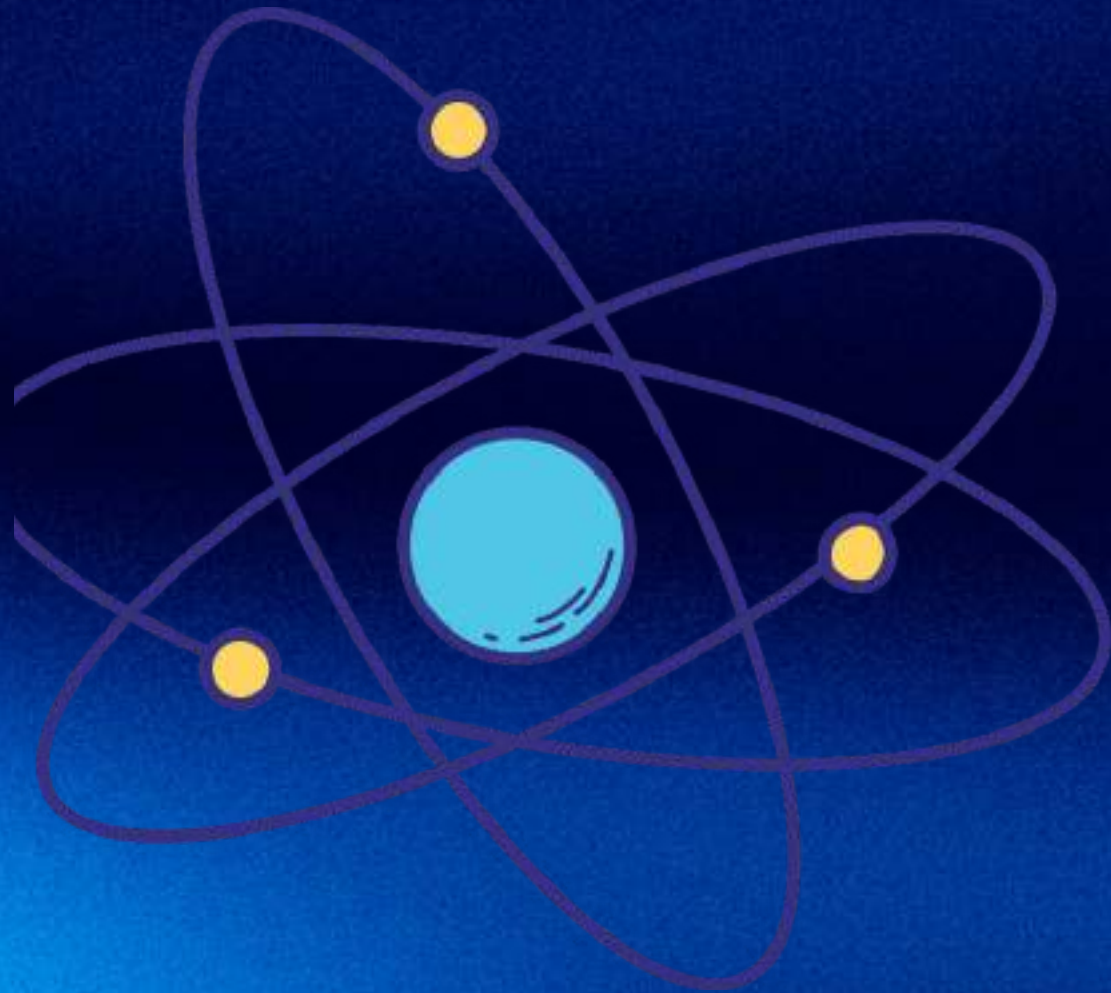
Data Structures & Big O

<u>Operation</u>	<u>Structure</u>	<u>Time</u>
Search by ID	Sorted Array (Binary Search)	$O(\log n)$
Sort Patients	Array (Selection Sort)	$O(n^2)$
Call Next Patient	Priority Queue	$O(n)$
Add Visit	Linked List	$O(1)$
Total Paid (Recursive)	Linked List	$O(n)$

- Waiting Room: Priority Queue
(built from scratch)
- Visit History: Linked List
(built from scratch)

⚛ Live Demo

Live demo will be shown during the presentation.



Choose:

```
Sort by (1) name (2) age: ID  NAME  AGE
103    Nour Ibrahim   19
101    Ahmed Hassan   34
```

Sorted in 1 comparisons [selection sort]

Choose:

#	PATIENT	CASE	ARRIVED
1	Nour Ibrahim	EMERGENCY	09:07 PM
2	Ahmed Hassan	NORMAL	09:07 PM

Choose:
Goodbye.